Restaurant Visitors Forecasting Using SARIMAX

📌 Project Overview

This project focuses on forecasting daily restaurant visitor demand using time-series forecasting techniques.

The project analyzes **478 daily restaurant visitor records** and develops **SARIMA and SARIMAX** forecasting models to capture temporal patterns and weekly seasonality.

The SARIMAX model additionally incorporates **holiday information as an exogenous variable** to improve forecasting performance.

The final model is used to generate a **7-day ahead restaurant visitor forecast** to support short-term staffing, inventory, and operational planning.

🎯 Problem Statement

Restaurants need accurate estimates of future customer demand to efficiently manage:

- Staff scheduling
- Food and inventory preparation
- Resource allocation
- Operational planning
- Customer service capacity

The objective of this project is to develop a time-series forecasting solution that can predict restaurant visitors based on historical visitor patterns and holiday information.

📊 Dataset

The dataset contains **478 daily observations** of restaurant visitor activity.

Important Features

Feature	Description
date	Date of the observation
weekday	Day of the week
holiday	Holiday indicator
total	Total number of restaurant visitors — target variable

The original dataset also contained restaurant-specific visitor columns and holiday names. These were not used in the final modeling process.

🔍 Exploratory Data Analysis

The project includes analysis of:

- Daily visitor trends
- Weekly seasonality
- Visitor patterns by weekday
- Holiday vs non-holiday visitor behavior
- Rolling statistics
- Time-series patterns
- Stationarity

These analyses were performed to understand the underlying structure of restaurant visitor demand before building forecasting models.

🧪 Time-Series Preprocessing

The following preprocessing steps were performed:

1. Converted the date column to datetime format.
2. Sorted observations chronologically.
3. Used the date as the time-series index.
4. Removed unnecessary columns.
5. Checked for missing values and duplicate records.
6. Split the data chronologically into training and testing sets.

Train-Test Split

- **Training observations:** 440
- **Testing observations:** 38

A chronological split was used instead of random splitting to preserve the temporal structure of the data.

📈 Stationarity Analysis

The Augmented Dickey-Fuller (ADF) test was used to evaluate the stationarity of the visitor time series.

Differencing was performed when required to make the series suitable for time-series modeling.

ACF and PACF plots were also analyzed to understand autocorrelation and help identify suitable model parameters.

🤖 Models Used

1. SARIMA

SARIMA was implemented to capture:

- Autoregressive patterns
- Moving-average patterns
- Differencing
- Weekly seasonality

The seasonal period was based on the observed **7-day weekly pattern**.

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2. SARIMAX

SARIMAX extends SARIMA by allowing external variables to influence the forecast.

In this project:

Holiday status was used as an exogenous variable.

This allows the model to account for changes in visitor demand associated with holidays.

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3. Auto-ARIMA

Auto-ARIMA was used to automatically search for suitable ARIMA/SARIMA parameters based on model performance.

The selected parameters were then used to implement the final forecasting model.

📊 Model Evaluation

The models were evaluated on the **38-observation test set** using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE)

SARIMAX Test Performance

Metric	Value
MAE	**18.81**

Metric	Value
RMSE	**23.37**
MAPE	**15.38%**

These metrics represent the model's performance on previously unseen observations.

🔮 7-Day Ahead Forecasting

After evaluating the forecasting model, the final SARIMAX model was retrained using the available historical data.

The final model was then used to forecast restaurant visitors for the **next 7 days**.

The forecast also includes confidence intervals to represent forecasting uncertainty.

Forecasted Visitors

Date	Predicted Visitors
2017-04-23	170.40
2017-04-24	79.73
2017-04-25	92.34
2017-04-26	96.11
2017-04-27	91.20
2017-04-28	141.88
2017-04-29	225.42

The forecast indicates variation in expected visitor demand throughout the week, with higher visitor demand predicted toward the weekend.

📉 Confidence Intervals

The forecasting model also generates confidence intervals around the predicted visitor counts.

For example:

Date	Forecast	Lower CI	Upper CI
2017-04-23	170.40	115.54	225.25
2017-04-24	79.73	24.88	134.59
2017-04-25	92.34	37.49	147.20

Date	Forecast	Lower CI	Upper CI
2017-04-26	96.11	41.25	150.96
2017-04-27	91.20	36.34	146.06
2017-04-28	141.88	87.02	196.73
2017-04-29	225.42	170.56	280.28

These intervals help communicate the uncertainty associated with each forecast.

💡 Business Insights

The forecasting solution can help restaurant management with:

🧑‍🍳 Staff Planning

Estimate expected customer demand and schedule an appropriate number of employees.

📦 Inventory Planning

Prepare sufficient food ingredients and supplies based on expected visitor volume.

💰 Resource Allocation

Allocate resources according to predicted demand.

📅 Short-Term Planning

Use the 7-day forecast to prepare for upcoming demand fluctuations.

🎉 Holiday Planning

Use holiday information to account for changes in visitor behavior.

🛠️ Technologies Used

- **Python**
- **Pandas**
- **NumPy**
- **Matplotlib**
- **Seaborn**
- **Statsmodels**
- **pmdarima**
- **Scikit-learn**
- **Jupyter Notebook**

Time-Series Techniques

- Time Series Analysis
- Stationarity Testing
- ADF Test
- Differencing
- ACF & PACF
- SARIMA
- SARIMAX
- Auto-ARIMA
- Seasonal Forecasting
- Confidence Intervals
- Model Evaluation

📁 Project Structure

Restaurant-Visitors-Forecasting/

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├── RestaurantVisitors.csv

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├── Restaurant_Visitors_Forecasting.ipynb

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└── README.md

🚀 Project Workflow

Dataset

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Data Cleaning

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DateTime Processing

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Exploratory Data Analysis

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Seasonality Analysis

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Stationarity Testing

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Differencing

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ACF / PACF

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Train-Test Split

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Auto-ARIMA

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SARIMA Model

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SARIMAX Model

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Model Evaluation

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Model Comparison

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Final SARIMAX Model

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7-Day Ahead Forecast

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Business Insights

📌 **Key Takeaways**

- Identified **weekly seasonality** in restaurant visitor demand.
- Used **Auto-ARIMA** for automated parameter selection.
- Built both **SARIMA and SARIMAX** forecasting models.
- Incorporated **holiday indicators as an exogenous variable**.

- Evaluated forecasting performance on a **38-observation test set**.
- Achieved **15.38% MAPE, 18.81 MAE, and 23.37 RMSE** with SARIMAX.
- Generated a **7-day ahead visitor forecast** with confidence intervals.
- Demonstrated how time-series forecasting can support restaurant operational planning.

🔮 Future Improvements

The model could be further improved by incorporating additional external factors such as:

- Weather conditions
- Restaurant promotions
- Special events
- Reservations
- Local events
- School or college holidays
- Seasonal festivals
- Customer booking data

These additional variables could potentially improve forecast accuracy and make the model more useful for real-world restaurant demand planning.

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